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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 01/07/2025 |
| **PI Name:** | Driskell, James Ph.D. | | |
| **Project Title:** | Dyads and Triads at 140 Million Miles: Factors Affecting Interpersonal Relations in Long-Duration Spaceflight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 32169-4817 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D |
| **Start Date:** | 01/06/2021 | **End Date:** | 01/06/2026 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 01/06/2026 per A. Beitman/NASA JSC (Ed., 9/30/24).<br>NOTE: End date changed to 01/05/2025 per A. Beitman/NASA JSC (Ed., 11/29/23). | | |
| **Key Personnel Changes/Previous PI:** | None | | |
| **COI Name (Institution):** | Driskell, Tripp Ph.D. ( Florida Maxima Corp )<br>Burke, Shawn Ph.D. ( University of Central Florida ) | | |
| **Grant/Contract No.:** | 80NSSC21K0457 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | September 2024 Update: Per NASA's Human Factors and Behavioral Performance Element, a one-year augmentation to this grant added an Antarctic analog and 1 year to the period of performance (POP). The Driskell team is augmenting their NASA research by utilizing the Amundsen-Scott, McMurdo, and Palmer stations as an analog for teams operating in long duration, isolated, confined environments (Ed., 9/30/24).<br>Psychosocial dynamics within the context of spaceflight have been shown to have a large impact on crew dynamics and well-being. As NASA moves towards longer duration exploration missions, the isolation and confinement may foster a blurring of work and social factors. The increased social support provided by crew members along with increased social interdependence has the potential to cause differential levels of interpersonal and/or intimate relationships among subsets of the crew. We propose a multi-faceted program of research to identify how trigger events will disrupt interpersonal relationships and corresponding markers of factors fostering social resilience. We will develop, test, and validate an individual-level and team-level countermeasure to facilitate social resilience in terms of accurate assessment of affect-based team dynamics and mitigate adverse effects of shifts in team dynamics due to interpersonal relationships. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Project deliverables will document: (1) factors associated with the development and breakdown of social resilience at the dyad and team levels across a range of interpersonal relationships, as well as characterization of the risk posed to team functioning and performance, (2) strategies and countermeasures to foster social resilience and mitigate the risk of dysfunctional interpersonal relationships that may threaten team coordination and cohesion, (3) the implementation and evaluation of a set of countermeasures with respect to their effectiveness, acceptability, and feasibility for use, and (4) recommendations and guidelines for the implementation of developed countermeasures within mission scenarios |
| **Task Progress:** | According to Landon, Vessey, and Barrett (2016), long-duration missions offer the possibility for the development of close dyadic relationships and potential subgroups within the team, and “research is needed to understand the effects of close relationships on team well-being and performance” (p. 52). Because of the increased interdependency within the team, increased isolation from family and friends, and prolonged exposure to mission demands, the risk exists that team members may form close or intimate dyadic or triadic relationships (e.g., subgroups or cliques) within the team that may disrupt team cohesion, cooperation, and coordination.<br>The goals of this research project are to (a) identify the risks to team functioning and performance stemming from the potential development of interpersonal relationships that may have adverse effects on the team, and (b) develop and test countermeasures to sustain team cohesion and mitigate the adverse effects of dysfunctional interpersonal relationships on the team.<br><br>To date, research has been conducted to (1) examine communication and interpersonal dynamics in analog spaceflight crews, (2) conduct structured interviews with prior astronauts to gain insight into crew dynamics in previous long-duration missions, (3) conduct experimental laboratory research to examine drivers of subgroup development within teams, and (4) design and develop countermeasures to mitigate adverse effects of dysfunctional crew dynamics. |
| **Bibliography Type:** | Description: (Last Updated: 01/09/2026) |
| **Abstracts for Journals and Proceedings** | Burke S, Kaosaar A, Szabó K, Driskell T, Driskell J. "The human side of long-duration space missions: A qualitative examination of astronauts’ interpersonal experiences." 2024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024.<br>Abstracts. 024 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 |
| **Abstracts for Journals and Proceedings** | Driskell T, Burke CS, Kaosaar A, Shorey A, Wilhelmy T, Driskell J. "Examining drivers and inhibitors of subgroup formation." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |
| **Abstracts for Journals and Proceedings** | Wilhelmy TA, Shorey A, Käosaar A, Burke CS, Driskell T, Driskell J. "Subgrouping in teams: Testing an intervention for mitigation." 2025 Society for Industrial and Organizational Psychology Annual Conference, Denver, Colorado, April 2-5, 2025.<br>2025 Society for Industrial and Organizational Psychology Annual Conference, Denver, Colorado, April 2-5, 2025. , Apr-2025 |
| **Articles in Peer-reviewed Journals** | Driskell T, Burke CS, Driskell JE. "Interpersonal dynamics in spaceflight: An archival study." Acta Astronaut. 2025 Apr;229:861-5. https://doi.org/10.1016/j.actaastro.2025.01.065 , Apr-2025 |
| **Articles in Peer-reviewed Journals** | Käosaar A, Burke CS. "The effect of challenge characteristics on teamwork and affective states in ICE." Group Organ Manag. 2025 Dec;50(5-6):1876-1931. https://doi.org/10.1177/10596011241304132 , Dec-2025 |